

Adham El-Messiry

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Professional Summary

Mechatronics Engineering student with experience in AI systems, robotics, embedded systems, and automotive diagnostics. Skilled in integrating software (Python, C++, MATLAB) with hardware through sensors, control logic, and real-time systems.

Experience includes deploying local LLMs, building AI pipelines, developing autonomous robotics, and performing diagnostics at AGMC BMW. Participated in the RTA Innovation Challenge developing an AI-based smart traffic system.

Education

Canadian University Dubai
BSc Mechatronics Engineering

Expected 2027

Experience

Rosa AI Startup — Jr. AI Service Engineer

Jul 2026 – Sep 2026

- Configured NVIDIA DGX Spark environments, validating OS, CUDA, and Docker under strict network policies.
- Deployed containerized local LLMs (vLLM/TensorRT) and verified end-to-end Application-to-Database pipelines.
- Benchmarked n8n workflows against hardware specs and documented on-premise AI integration standards.
- Built an automated customer onboarding pipeline for web scraping, persona validation, and PostgreSQL storage.
- Designed backend logic to dynamically customize an AI sales assistant ("Bud") using extracted client data.

AGMC BMW — Engineering Intern

Sep 2025 – Nov 2025

- Performed diagnostics using BMW ISTA and CAN Bus systems.
- Assisted in mechanical and electrical repairs and calibration.
- Conducted system inspections and testing.

Maktoub — Founder

2025 – Present

- Built clothing brand including product design, sourcing, and sales.
- Managed branding, marketing, and digital presence.

Competition

RTA Innovation Challenge — AI Smart Traffic System

2026

- Developed AI-based system to prioritize emergency vehicles and reduce congestion.
- Built live demo simulating real-time traffic scenarios.
- Used Mistral AI and applied Responsible AI principles.
- Contributed to business, marketing, and data strategy.

Projects

AI Autonomous Robocar

- Built Raspberry Pi vehicle with camera and ultrasonic sensors.
- Implemented real-time lane detection and navigation.

Smart Street Lighting

- Designed adaptive system using LDR and motion sensors.
- Achieved 35% simulated energy reduction.

Motorcycle Restoration

- Rebuilt mechanical components and full electrical wiring.
- Diagnosed faults using systematic testing.

Technical Skills

- **Programming:** Python, C/C++, MATLAB, Arduino Programming, Structured Text (PLC)
- **AI/Tools:** Machine Learning, Generative AI, Object Detection, OpenCV, Image Processing, Mistral AI
- **Systems:** Arduino, Raspberry Pi, Arduino UNO Q, Sensors, Ultrasonic Sensors, LDR Sensors, CAN Bus, ECU Diagnostics, Docker, PostgreSQL, n8n
- **Hardware:** PI Control, Feedback Control Systems, PLC, Pneumatics, Hydraulics, Process Control
- **Software:** 3D CAD Modelling, Mechanical Design, Stress Analysis, Manufacturing Processes

Certifications

RTA Innovation Challenge Certificate

Arduino UNO Q Workshop - Besomi Academy

Certificate of Appreciation - Canadian University Dubai

Apply AI: Analyze Customer Reviews - Cisco Networking Academy

Introduction to Internet of Things (IoT) - Cisco Networking Academy

Introduction to Modern AI - Cisco Networking Academy

Generative AI Explained - NVIDIA